

Spanning Tree Protocol (STP) Lab Guide — Cisco Packet Tracer

Topology Overview

Switch	Connects to	Local Interface	Remote Interface
SW1	SW2	G0/2	G0/1
SW1	SW3	G0/1	G0/2
SW2	SW3	G0/2	G0/1

Switch	PCs Attached	VLAN
SW1	PC1 (10.0.0.1/24), PC2 (10.0.0.2/24)	VLAN 10
SW1	PC3 (10.0.0.3/24)	VLAN 30
SW2	PC4, PC5, PC6 (10.0.0.4-6/24)	VLAN 20
SW3	PC7, PC8 (10.0.0.7-8/24)	VLAN 30
SW3	PC9 (10.0.0.9/24)	VLAN 20

The three switches form a **triangle (loop)** — this is what makes STP necessary.

Part 0 — Build the Topology in Packet Tracer

1. Place 3× **2960 switches** (SW1, SW2, SW3) and 9× PCs as shown in the diagram.
2. Cable the switches together with **copper straight-through** cables in a triangle:
 - SW1 G0/2 ↔ SW2 G0/1
 - SW1 G0/1 ↔ SW3 G0/2
 - SW2 G0/2 ↔ SW3 G0/1
3. Cable each PC to its switch using the FastEthernet ports shown in the diagram.
4. Assign the IP addresses shown above to each PC (Desktop > IP Configuration), using subnet mask 255.255.255.0.
5. On each switch, create the VLANs and assign access ports:

```
enable
configure terminal
vlan 10
  name VLAN10
vlan 20
  name VLAN20
vlan 30
  name VLAN30
exit
```

Then assign each PC-facing port to the correct VLAN, e.g. on SW1:

```
interface fastEthernet 0/1
  switchport mode access
  switchport access vlan 10
interface fastEthernet 0/2
  switchport mode access
  switchport access vlan 10
interface fastEthernet 0/3
  switchport mode access
  switchport access vlan 30
```

Repeat on SW2 and SW3 for their respective PCs/VLANs.

6. Configure the three inter-switch links (G0/1, G0/2 on each switch) as **trunk** ports so all VLANs can traverse them:

```
interface gigabitEthernet 0/1
  switchport mode trunk
interface gigabitEthernet 0/2
  switchport mode trunk
```

Do this on all three switches, on the two Gig ports that connect to the other switches.

7. Verify connectivity within each VLAN using (ping) from the PC command prompt (PCs in the same VLAN should reach each other; different VLANs will not, since no routing is configured — that's expected for this lab).
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Part 1 — Examine the Default STP Behavior

Goal: Answer the questions from the lab sheet using `show` commands — do this on **all three switches**.

1. Check the current STP version:

```
show spanning-tree summary
```

By default, Cisco switches run **PVST+ (Per-VLAN Spanning Tree Plus)**, the Cisco-proprietary version of 802.1D STP. This is what you should observe unless someone has changed it.

2. Check the Bridge ID and Root Bridge for each VLAN:

```
show spanning-tree vlan 10
show spanning-tree vlan 20
show spanning-tree vlan 30
```

For each VLAN, note:

- **Bridge ID** of the local switch (priority + MAC address) — look at the "Bridge ID" section.
- **Root ID** — this tells you which switch is the root bridge for that VLAN (compare the Root ID's MAC address to each switch's own MAC to identify which physical switch it is).
- Since no priority has been manually configured yet, all switches start with the **default priority of 32768**. The root bridge is therefore whichever switch has the **lowest MAC address** for that VLAN.

3. Check interface roles and costs:

```
show spanning-tree vlan 10 detail
```

or simply read the table from `show spanning-tree vlan 10`. Note for each interface:

- **Role:** Root (R), Designated (D), Alternate/Blocking (A/B)
- **Cost:** the default STP cost is based on bandwidth — a Gigabit link has a default cost of 4 (100 Mbps link would show cost 19).
- **State:** Forwarding or Blocking.

4. Identify the **blocked port**: In a 3-switch triangle, one port on the non-root switches will always end up in **Blocking** state to break the physical loop. Find it in the output and note:
- Which switch and interface is blocked
 - **Why**: it's the port that has the worst (highest) root path cost, or on a tie, the one connected to the switch/port with the higher Bridge ID/Port ID — STP blocks it to prevent a Layer 2 loop while still keeping the link available as a backup.

Tip: run these three `show spanning-tree vlan X` commands on **each of the 3 switches** and record the results in a table — this is exactly what the lab sheet is asking for.

Part 2 — Change STP Mode to Rapid-PVST

Rapid-PVST (802.1w) converges much faster than legacy PVST+ (802.1D).

On **each switch** (SW1, SW2, SW3):

```
enable
configure terminal
spanning-tree mode rapid-pvst
end
```

Verify the change:

```
show spanning-tree summary
```

You should now see the STP mode listed as **rapid-pvst** instead of the legacy default.

Part 3 — Configure Root Bridge Priorities per VLAN

Requirement:

VLAN	Primary Root	Secondary Root
10	SW1	SW2
20	SW2	SW3
30	SW3	SW1

Use the `(root primary)`/`(root secondary)` macro commands — Packet Tracer/IOS automatically sets a low enough priority (24576 for primary, 28672 for secondary, or lower still if needed to beat existing bridges).

On SW1:

```
configure terminal
spanning-tree vlan 10 root primary
spanning-tree vlan 30 root secondary
end
```

On SW2:

```
configure terminal
spanning-tree vlan 20 root primary
spanning-tree vlan 10 root secondary
end
```

On SW3:

```
configure terminal
spanning-tree vlan 30 root primary
spanning-tree vlan 20 root secondary
end
```

Verify on each switch:

```
show spanning-tree vlan 10
show spanning-tree vlan 20
show spanning-tree vlan 30
```

Confirm the "This bridge is the root" message appears under the correct VLAN on the correct switch, and check the priority values (e.g., 24576 for root primary).

Alternative manual method (if you want explicit control instead of the macro):

```
spanning-tree vlan 10 priority 4096
```

Priority must be a multiple of 4096 (0, 4096, 8192, ... 61440).

Part 4 — Enable PortFast and BPDU Guard

Important: PortFast and BPDU Guard should only be enabled on **access ports connecting to end devices (PCs)** — never on trunk ports between switches, since those are legitimate paths for BPDUs and loops.

On each switch, go into the interfaces facing PCs (e.g., on SW1: Fa0/1, Fa0/2, Fa0/3):

```
configure terminal
interface range fastEthernet 0/1 - 3
    spanning-tree portfast
    spanning-tree bpduguard enable
end
```

Repeat with the correct interface ranges on SW2 and SW3 for their PC-facing ports.

Verify:

```
show spanning-tree summary
show running-config interface fastEthernet 0/1
```

You should see `spanning-tree portfast` and `spanning-tree bpduguard enable` under each access interface, and the summary should list the number of ports with PortFast enabled.

Why this matters:

- **PortFast** skips the Listening/Learning delay (normally ~30-50 sec) and puts the access port straight into Forwarding, so PCs get network access immediately after boot instead of waiting.
- **BPDU Guard** protects against a loop or rogue switch being plugged into that access port: if a BPDU is ever received on a PortFast-enabled port, the port is immediately put into **err-disabled** state, shutting it down rather than allowing a loop to form.

Final Verification Checklist

- ☐ Topology cabled correctly (triangle between SW1/SW2/SW3, PCs attached per diagram)
- ☐ VLANs 10, 20, 30 created and PC ports assigned correctly
- ☐ Inter-switch links configured as trunks
- ☐ Recorded default STP version, Bridge IDs, root bridges, interface costs, and blocked port for each switch/VLAN
- ☐ STP mode changed to `rapid-pvst` on all three switches

☐ Root/Secondary root priorities configured per the VLAN table above and confirmed with

`show spanning-tree vlan X`

☐ PortFast + BPDU Guard enabled only on PC-facing access ports, confirmed with `show`

`running-config` or `show spanning-tree summary`

Useful commands to keep handy throughout:

```
show spanning-tree summary
show spanning-tree vlan <id>
show spanning-tree vlan <id> detail
show spanning-tree interface <intf> detail
show running-config
```